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Review

Collective action impacts on climate change mitigation

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To address climate change, we need immediate and sustained collective action at multiple scales. In this short, narrative review, we examine recent research on collective action to mitigate climate change (referred to as collective climate action). We explore the conceptualization of climate change as a collective action problem and review documented interventions and outcomes. Although researchers have made progress in understanding what drives collective climate action, gaps remain in evaluating practical interventions and measuring long-term impacts. Opportunities exist for future interdisciplinary research to bridge individual and collective approaches, improve outcome measurement, and design interventions that can more effectively promote collective action to address climate change.

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Introduction

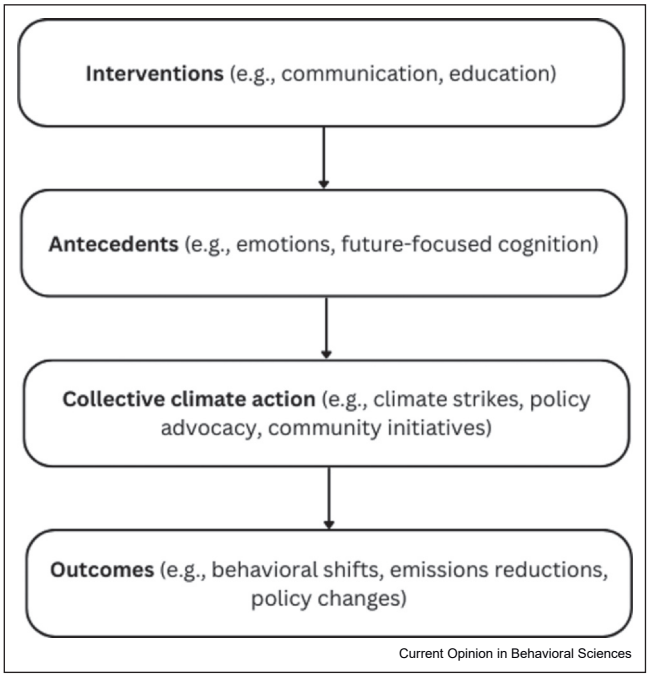
Achieving the needed reductions in greenhouse gas emissions to limit global warming requires transformative changes in worldwide systems and structures [1]. Efforts to reduce greenhouse gas emissions, referred to as climate change mitigation, include transitioning to renewable energy sources, improving energy efficiency, enhancing carbon sinks, changing land-use practices, promoting sustainable consumption patterns and lifestyle choices, and developing low-carbon, carbon capture, and carbon storage technology. Recognizing the level of needed change given the inherent complexity of climate change and the failure of individual behaviors to meaningfully move the needle when it comes to greenhouse gas emissions, scholars, activists, and policymakers have called for collective action.

We define collective action as people working together to achieve a common goal. Collective action to mitigate climate change, which we term 'collective climate action', can take many forms from organized protests, such as climate marches, to people working together to pass renewable energy policies. Figure 1 shows the pathway from interventions through antecedents and collective action to outcomes. Although the study of collective action has a long history, we focus this review on recently published work that has (1) conceptualized climate change as a collective action problem, (2) described interventions to spur collective climate action, and (3) documented outcomes of collective climate action. Throughout the review, we draw on research and thought from many disciplines, recognizing that interweaving interdisciplinary perspectives can be particularly powerful as we address a multifaceted challenge such as climate change.

Review methods

Using the search terms of 'collective action' and 'climate change mitigation', we systematically searched the following academic databases: Academic Search Premier, Environment Index, GreenFILE, PsychINFO, and Web of Science. We limited our search to peer-reviewed sources published after 2022 (the search was conducted in September 2024). We imported 563 citations into Covidence, a research review management software program, which then removed 62 duplicates. Additionally, we imported 31 citations found through various means of serendipitous searching, including Google Scholar searches, personal knowledge, and colleague recommendations. We screened abstracts of the 533 citations, including only citations that substantially focused on climate change *and* collective action. Finally, we retrieved the full text for the included 176 citations and imported them into NVivo, a qualitative analysis program, where we coded for emergent themes. We selected final articles based on (1) strength of empirical evidence or theoretical contribution and (2) relevance to the review topic and identified themes. When multiple papers addressed similar findings within a theme, we prioritized more recent publications and those offering interdisciplinary perspectives.

Figure 1



Model showing the collective climate action process, from interventions through antecedents and collective action to outcomes.

From our thematic coding, several themes emerged that structure the following discussion of collective climate action research. First, we explore how climate change is increasingly conceptualized as a collective action problem, requiring coordinated responses across multiple scales. Second, we examine interventions designed to motivate collective climate action. Third, we categorize and discuss documented outcomes of collective climate action. Our review and coding also revealed five trends examining specific topics and their intersection with collective climate action: young people [2–10], climate justice [7,8,11,12], social media [13–18], COVID-19 [5,19–21], and health [2,5,19–26]. These topical trends reflect current events, can help researchers understand the state of the field, and identify areas for cross-disciplinary integration. In-depth discussion of each trend is beyond the scope of this review, as we focus primarily on three core themes regarding conceptualization, interventions, and outcomes of collective climate action.

Climate change as a collective action problem

Despite efforts of some corporations, industries, and politicians to frame climate change as an individual problem (thereby dodging responsibility), scholars and communities increasingly recognize climate change as a collective issue [27]. Climate change is diffused with collectiveness [28] as it is caused by the sum of individual actions (primarily the burning of fossil fuels but also destroying carbon sinks). The aggregate of these actions creates unintended (but somewhat predictable) consequences (e.g. changes in climate that cause harm, such as frequent and more intense weather events, sea-level rise, ocean acidification, and changes in precipitation, among others) that cannot be addressed by individuals acting alone. When considering how to meaningfully mitigate climate change, the aggregated, complex, and dispersed causes dictate the need for action by collective entities [28]. Climate change mitigation is further complicated by some individuals benefiting from fossil fuel use at the expense of others (free-riding); a potential lack of incentive for current generations to care about future generations; and a lack of a willing and capable entity to force a move away from fossil fuels [29]. Although individual actions to reduce carbon footprints are important, they alone cannot address the scale and complexity of climate change. The problem's collective nature necessitates collective climate action to raise awareness and motivation at meaningful levels, create comprehensive policies and regulations, and build the social and institutional structures needed for widespread emissions reductions.

The required collective climate action can take many forms and involve various actors across scales. At an institutional level, collective climate action includes collaborative governance [30], carbon offset programs [31], and investor coalitions [32]. At a community level, it involves groups working together on local climate initiatives [33]. When examining collective action from an individual's perspective, researchers often classify actions as normative or non-normative [34–37]. Normative collective actions align with established social norms and can include, for example, in a U.S. context, volunteering, donating money, petitioning, voting, lobbying, and joining peaceful protests. Non-normative collective actions violate social norms and may include vandalism, violence, or other illegal activities. Recent research includes an emphasis on climate activism, with many researchers studying global social movements such as the Fridays for Future climate strikes [35,38,39].

Emerging from the complexity and scale of climate change is an interdisciplinary conversation that invokes various models, frameworks, conceptualizations, and theories to understand how collective climate action develops and works, as well as its associated outcomes. These approaches include social-ecological framing, the Commons framework, social movement theory, and social identity models (see Table 1) [22]. Concurrently, researchers recognize the limitations of applying existing collective action theories and frameworks without modification; for example, some argue for rethinking common-pool resource theory to more effectively address large-scale environmental challenges like climate change [44] and call to expand thinking about collective

Table 1

Theoretical frameworks used to study collective climate action: key examples with associated actors and outcomes.

Theory/framework	Example article	Actors of focus	Collective action outcomes
Social-Ecological Framing emphasizes how environmental outcomes are shaped by the interconnectedness of social (people) and ecological (nature) systems.	Djinlev & Pearce, 2024 [24]	Communities (energy communities)	Behavioral and social outcomes (changes in social norms and structures)
	Kashima et al., 2023 [40]	Multilevel stakeholders (individuals, groups, organizations, governments)	Behavioral and social outcomes (development of shared mental models)
The Commons Framework focuses on how communities collectively manage shared resources without depleting them, emphasizing collective governance and institutional arrangements.	Agrawal et al., 2023 [41*]	Resource users (individuals, institutions managing common pool resources)	Policy and regulatory changes (changes in natural resource policy and management)
	Colding et al., 2022 [42]	Urban communities (urban green commons, coworking spaces, community climate commons)	Behavioral and social outcomes (environmental learning), emotional and psychological outcomes (empowerment), physical and environmental changes (reduced emissions)
Social Movement Theory explores how and why people engage in collective action for social change, emphasizing mobilization processes, issue framing, resource allocation, and political opportunities.	Mullen & Widener, 2022 [11]	Local stakeholders (community organizers, residents)	Behavioral and social outcomes (understanding and framings of local risks and disparities)
	Jimenez-Luque, 2024 [13]	Social movement participants (Buen Vivir movement members)	Behavioral and social outcomes (challenging the status quo)
Social Identity Models examine how group membership and shared identities impact collective action by shaping individual attitudes and social norms.	Ucar et al., 2023 [43]	Identity-based groups (national groups, environmentalists)	Behavioral and social outcomes (changes in group norms and beliefs)
	Romano et al., 2024 [3]	Youth activists (pro-environmental movement participants)	Emotional and psychological outcomes (improved mental health and well-being)

action to include an institutional focus [45]. Although sources cited thus far emphasize collective action, there are arguments (and counterarguments) downplaying the role of collective action [46–48], as well as calls for integrating individual actions (often in the domain of psychology and related fields) with collective action to more accurately reflect a dynamic, systems-level perspective [49–51].

Efforts to spur collective climate action: antecedents and interventions

Building on the premise that climate change is a collective action problem, recent research examines how we can encourage people to take collective climate action. Given that individual behaviors may be easier to motivate and sustain, researchers have investigated potential spillover effects from individual behaviors to collective action; however, evidence for the efficacy and persistence of such spillover effects remains limited [52]. As such, other researchers focus on identifying antecedents of collective climate action and developing interventions that can increase involvement in collective climate action. Scholarship on antecedents represents a large field of study beyond the scope of this review, but we note two categories of antecedents receiving increased scholarly interest: emotions (positive and negative) [3,35,37,53–59] and future-focused cognition [55,56,60–63].

Many of the interventions in our review focus on climate change communications, exploring various aspects of how information about climate change is shared and discussed, and how this influences participation in collective climate action. These communications vary in their approaches, with one study examining the impact of positive versus negative messaging [64] and another testing generational framing [4]. Researchers also investigated different formats for climate change communications, from climate fiction videos [65] to a climate-themed video-based intervention [66]. Several studies focused on relational communications, examining dialog spaces at the United Nations climate change conferences of the parties (COPs) [67], youth-led environmental communication with parents [68], and conversations between climate activists and nonactivists [69].

The interventions often aimed to motivate collective action by addressing various antecedents of collective climate action, with a focus on psychological factors. Several studies examined worry as a motivator, both among young people [4] and in parent–child relationships [68]. Others investigated a broader range of psychological antecedents of collective climate action, with one study examining 11 different theorized psychological predictors [66], while others focused on efficacy [69] or emotional arousal [64]. Beyond psychological factors, communication-focused interventions also sought to promote community building [4,67,69] and learning [65,67].

Although communication-focused interventions dominated our sample, education also emerged as a pathway for

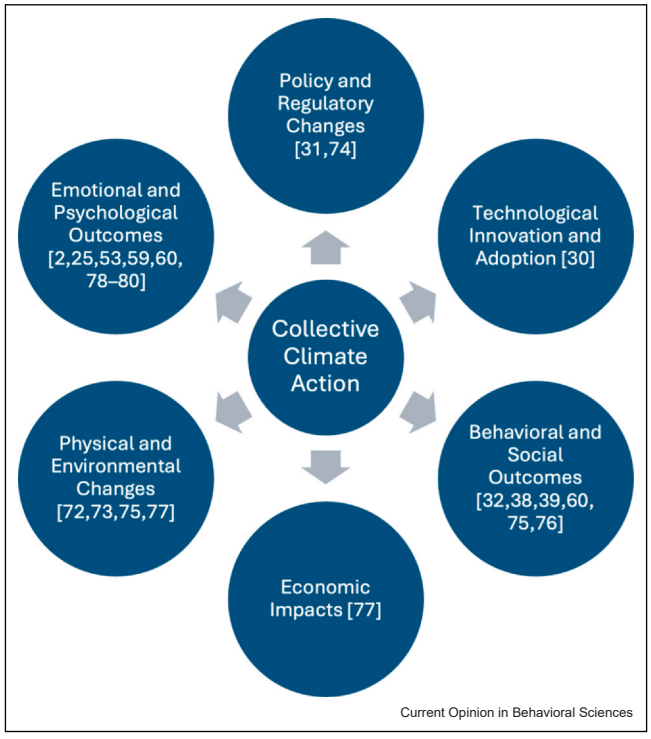
promoting collective climate action. One study acknowledged the critical role of education in supporting communication that can lead to collective climate action [68]. The researchers specifically identified how climate change education increases children's climate change knowledge, which leads to more conversations with their parents about climate change. Other educational interventions included simulation-based learning [17] and a school-based education for sustainable development program [2]. In a discussion paper, one group of scholars described their work, in collaboration with the Organisation for Economic Co-operation and Development (OECD), which focuses on developing the construct of 'Agency in the Anthropocene' to equip young people with the necessary competencies to collectively address complex socioecological challenges [70].

Outcomes of collective climate action

Given the collective nature of climate change [28], outcomes of collective action are often assumed, evidenced by the varied and numerous calls for collective climate action [23,71]. While assumptions about outcomes of collective action abound, a much smaller evidence base exists to document impacts. As part of this review, we sought to include recent literature that discussed and documented collective climate action outcomes; we identified 20 studies that did so (see Figure 2). Initially, we attempted to classify the outcomes in each study into seven categories of collective *environmental* action outcomes that had been identified in a previous review [34]: corporate/business, political, movement building, development projects, physical changes, change in public attitudes/values/behaviors, and legal. The outcomes in the 20 studies did not, however, map well with those outcomes, so we revised the categories to develop a list that more accurately reflected the collective climate action outcomes as reported in the 20 studies. Our coded categories of collective climate action are:

- *Emotional and psychological outcomes:* increases in positive emotions such as hope, optimism, and sense of empowerment; the mitigation of negative emotions such as anxiety and despair; and improvements in human well-being and mental health.
- *Behavioral and social outcomes:* changes in individual and group engagement in climate-related behaviors; increased awareness and concern about climate change; and movement building such as growth in organizational membership and strengthened activist networks.
- *Physical and environmental changes:* reductions in greenhouse gas emissions, creation of carbon sinks, and increases in renewable energy infrastructure.
- *Policy and regulatory changes:* adoption of climate policies and regulatory frameworks at local, national, and international levels.
- *Economic impacts:* divestment from fossil fuel industries, growth of green jobs, transition of workers away from carbon-intensive jobs, and growth of local economies resulting from climate action.
- *Technological innovation and adoption:* development and implementation of clean energy technology; infrastructure and adoption of energy-efficient technologies.

Figure 2



Outcomes of collective climate action with example studies providing evidence of positive outcomes [2,25,30–32,38,39,53,59,60,72–80].

Recent research on collective climate action provides evidence of outcomes, as illustrated in Figure 2, with emotional and psychological outcomes emerging as the most frequently documented category in our review. Figure 2 displays citations for studies focusing on *positive* outcomes, but negative or null outcomes of collective climate action also have been reported. In one study [54], for example, researchers shared survey data suggesting people who participated in collective climate action had higher levels of psychological distress (measured as stress, anxiety, and depression). In a comparative case study [72], a researcher shared two cases involving collectives working to convert pastures for cattle ranching to silvopastures and reported success for one group but not the other. Additionally, some research points to the need for trade-offs among different types of outcomes. For instance, using scenario simulation analysis, researchers discussed how carbon management via a collective action option was more effective but also more costly than a marginal opt-out strategy [73].

Conclusion

As the urgency of addressing climate change grows, so does the need for effective mitigation measures, especially through collective action. In this review, we synthesize recent research conceptualizing climate change as a collective action problem and document evidence of collective climate action outcomes across multiple dimensions — from emotional and psychological impacts to policy changes. From diverse drivers, barriers, and interventions to various outcomes and emerging trends, collective climate action is a complex and evolving field of study. Research demonstrates the importance of considering multiple levels of analysis, from individual psychological factors to broader societal and institutional contexts. The identified trends suggest promising directions for future research and highlight the interconnected nature of climate change with other societal issues. More scholarship is needed to bridge the gap between individual and collective approaches, explore innovative methods for measuring outcomes, and develop interventions that can scale effective collective action strategies.

Declaration of Competing Interest

There are no interests to disclose.

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